



# VLSI Testing

## 積體電路測試

### *Design For Testability*

### *Part 2: External Scan (JTAG)*

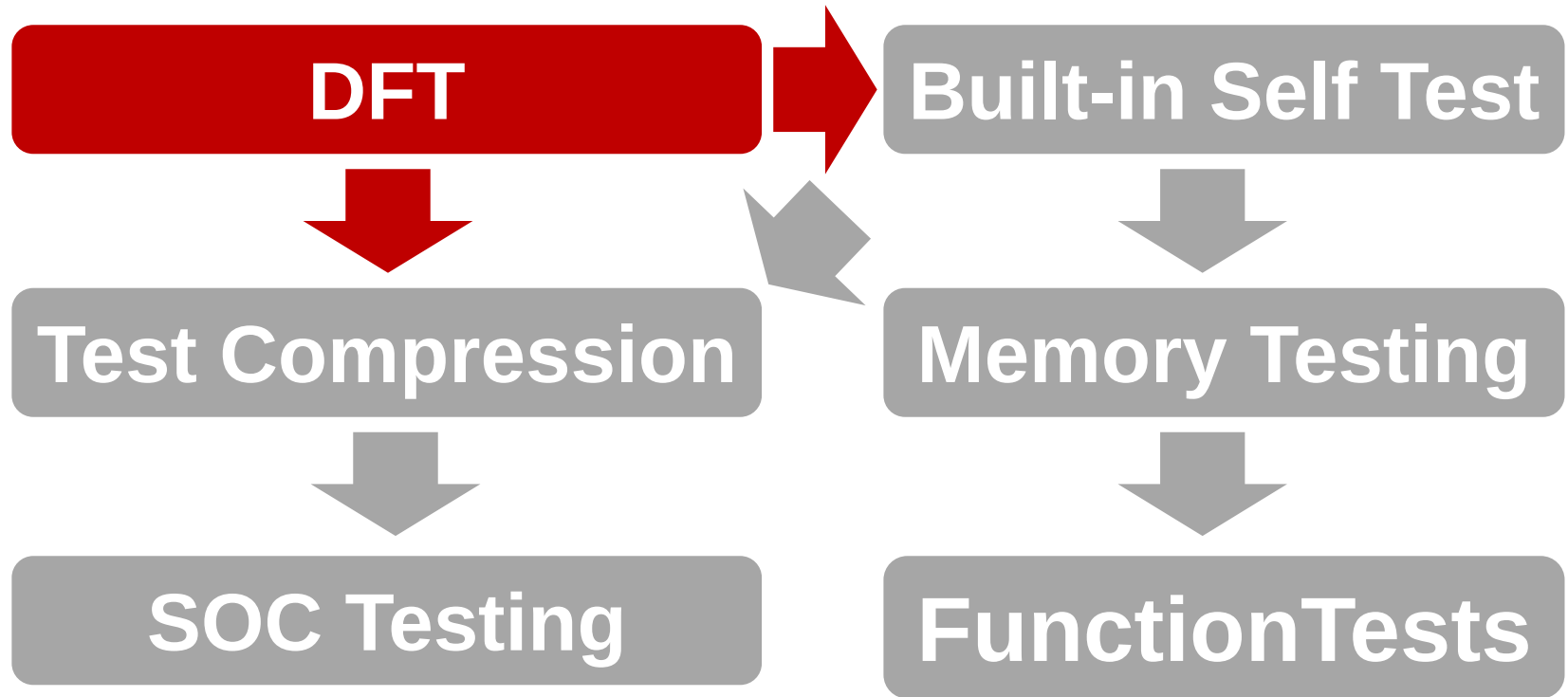
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**Graduate Institute of Electronics Engineering**

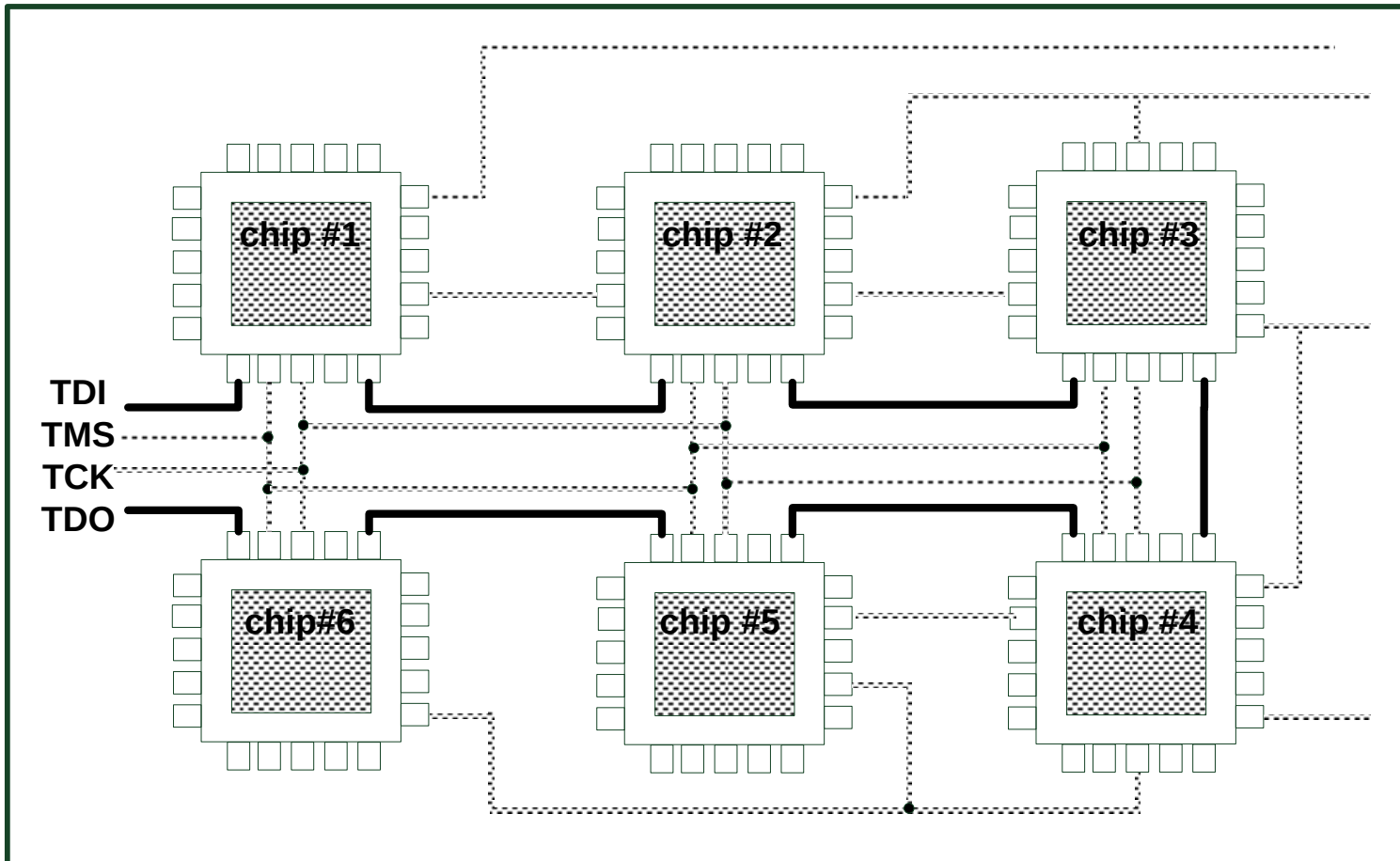
**National Taiwan University**

# Course Roadmap (Design Topics)



# Motivating Problem

- You buy 6 chips and assemble a board
  - ◆ Your manager asks you to test the board and chips, how?



# Why Am I Learning This?

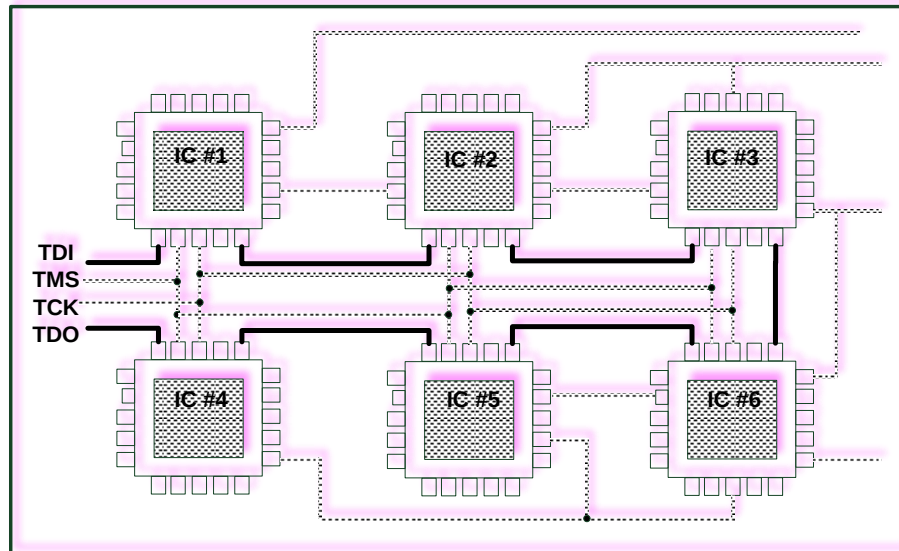
- This chapter allows us to understand
  - ♦ JTAG, IEEE 1149.1 standard
  - ♦ board-level testing

*“In school we learn things then take the test,  
In life we take the test then learn things.”*

*(Admon Israel)*

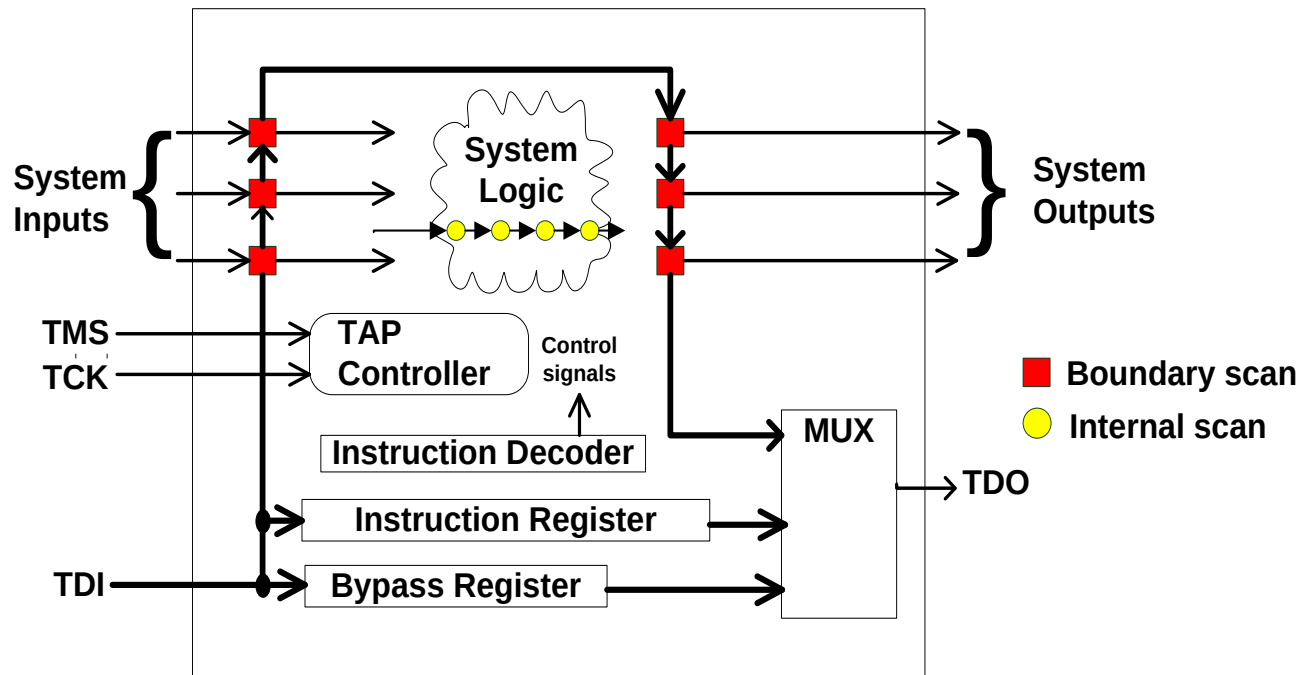
# DFT – Part 2

- Introduction
- JTAG Architecture and Components
- JTAG Instructions
- Conclusion



# What is External Scan ?

- **External scan**
  - ♦ Stitch **system input/output pins** into a shift register
    - \* as opposed to *internal scan*
  - ♦ also known as **boundary scan**

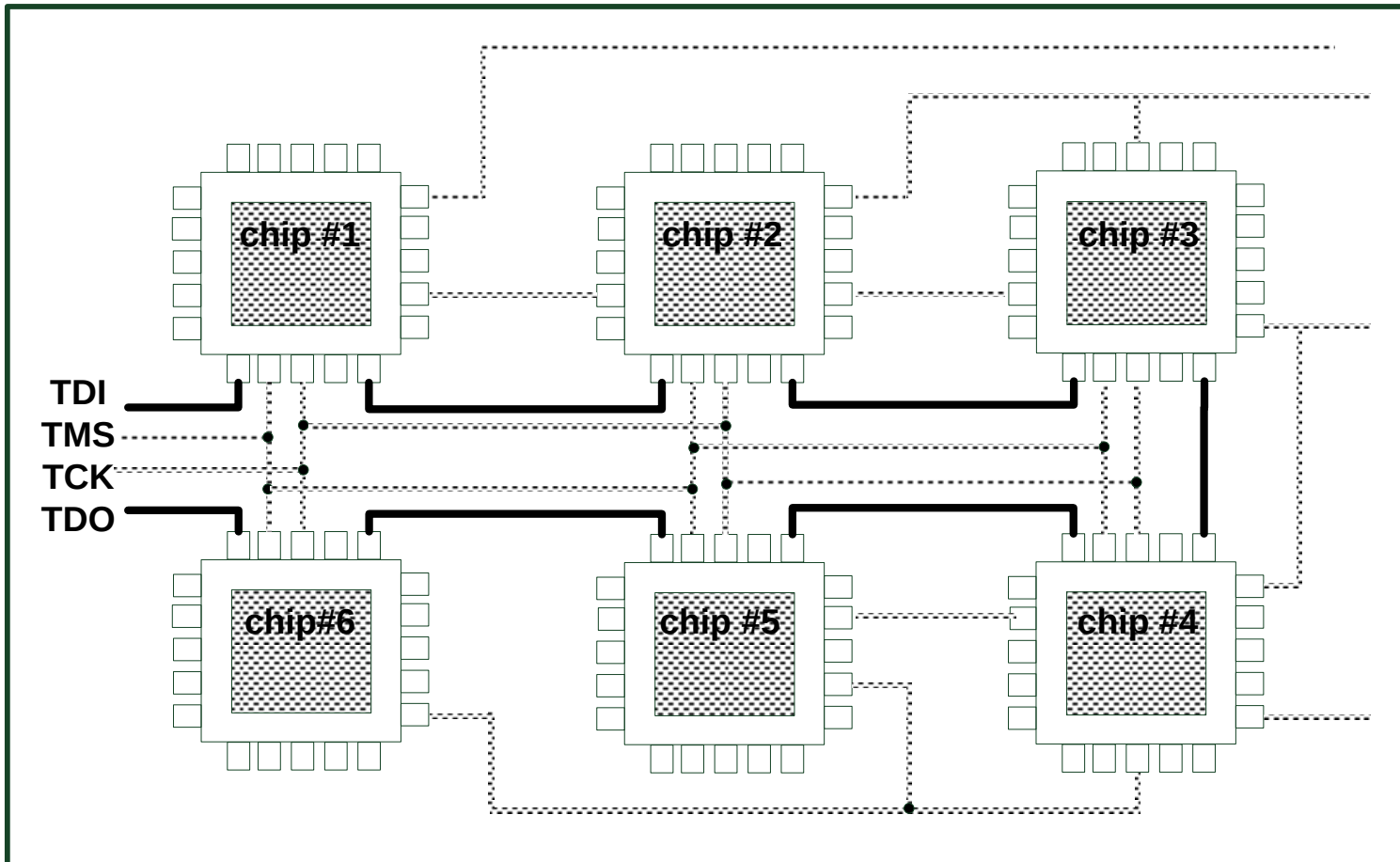


# Boundary Scan

- Boundary scan standard is needed
  - ♦ Allow chips from **different vendors** to be tested together
- **IEEE 1149.1-1990**
  - ♦ **Boundary scan** standard
  - ♦ aka **JTAG**
- Why use boundary scan?
  - ♦ 1. **Board-level** test and diagnosis
  - ♦ 2. Test on-board **interconnect** among chips
  - ♦ 3. Test on-chip **system logic**

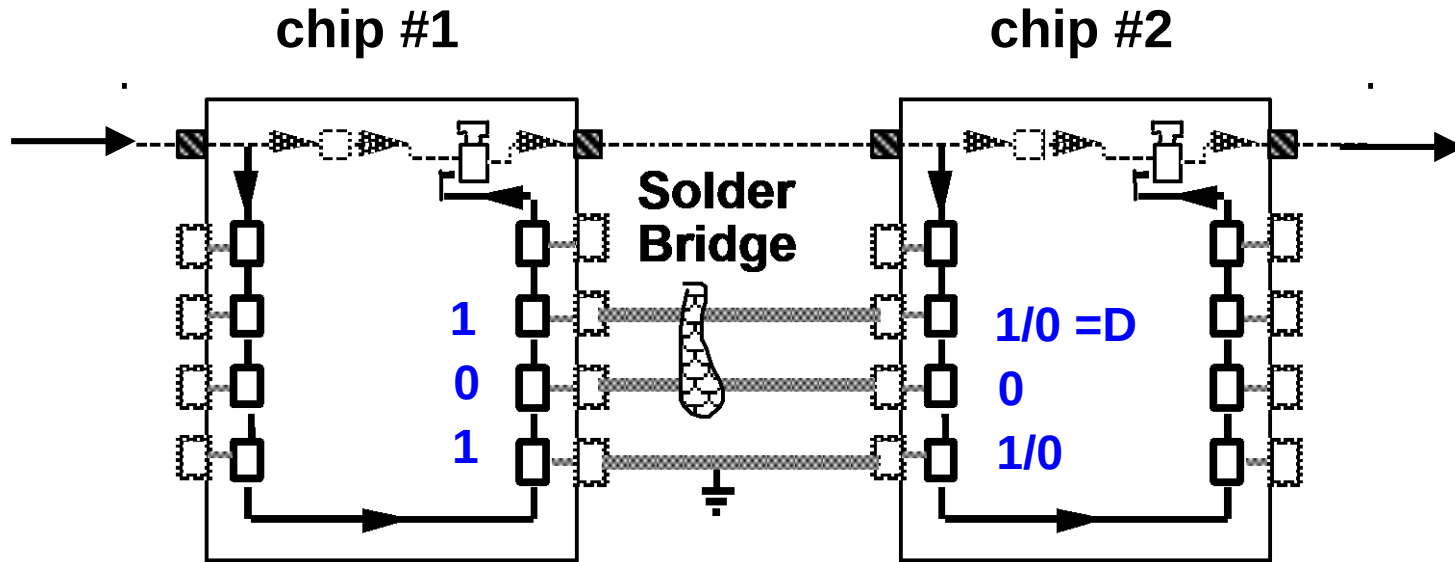
# Board Level Test and Diagnosis

- All chips are stitched into **one JTAG scan chain**
- Off-line testing and on-line debug are supported





# Test On-board Wires Among Chips

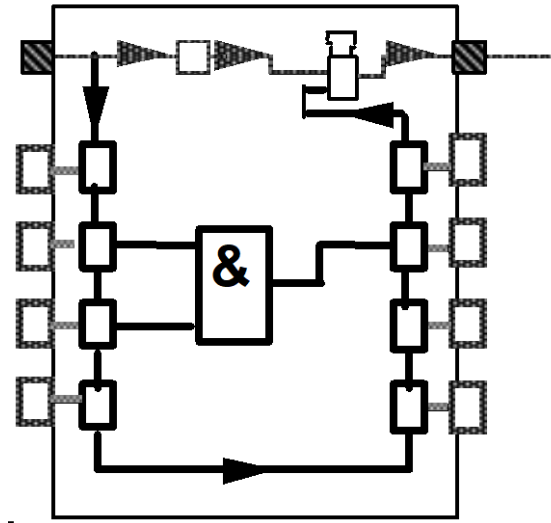


Input:           X X X X    1 0 1 X X X X X    X X X X ←First scan in  
 Output:        X X X X    X X X X X D 0 D    X X X X ←First scan out

\*assume wired-AND fault model

# Test On-chip System Logic

- How to test a chip already assembled on board. Cannot use ATE



|        |   |   |   |   |   |   |   | First scan out |         |   |   |   |   |   |   |   |   |
|--------|---|---|---|---|---|---|---|----------------|---------|---|---|---|---|---|---|---|---|
|        |   |   |   |   |   |   |   | First scan in  |         |   |   |   |   |   |   |   |   |
|        |   |   |   |   |   |   | ↓ |                |         |   |   |   |   |   |   | ↓ |   |
| Input: | X | 1 | 1 | X | X | X | X | Good           | Output: | X | X | X | X | X | X | 1 | X |
| Input: | X | 0 | 1 | X | X | X | X | Good           | Output: | X | X | X | X | X | X | 0 | X |
| Input: | X | 1 | 0 | X | X | X | X | Good           | Output: | X | X | X | X | X | X | 0 | X |

# Quiz

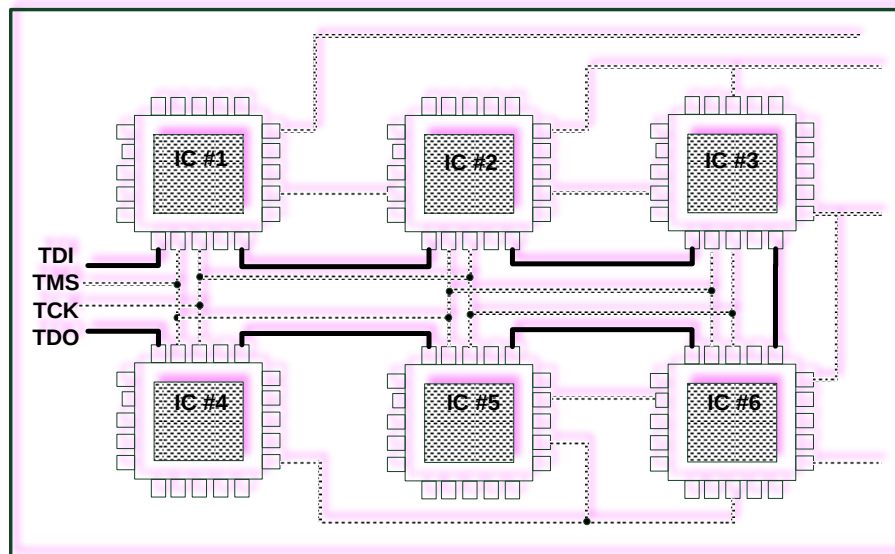
**Q: Which of the following is NOT true about boundary scan?**

- A. Boundary scan enables board-level testing**
- B. Requires standard because chips are from different vendors**
- C. Boundary scan can replace internal scan**

**ANS:**

# DFT – Part 2

- Introduction
- JTAG Architecture and Components
  - ◆ TAP
  - ◆ TAP controller
  - ◆ Registers
  - ◆ Instruction Decoder
- JTAG Instructions
- Conclusion



# JTAG Architecture

- JTAG Components

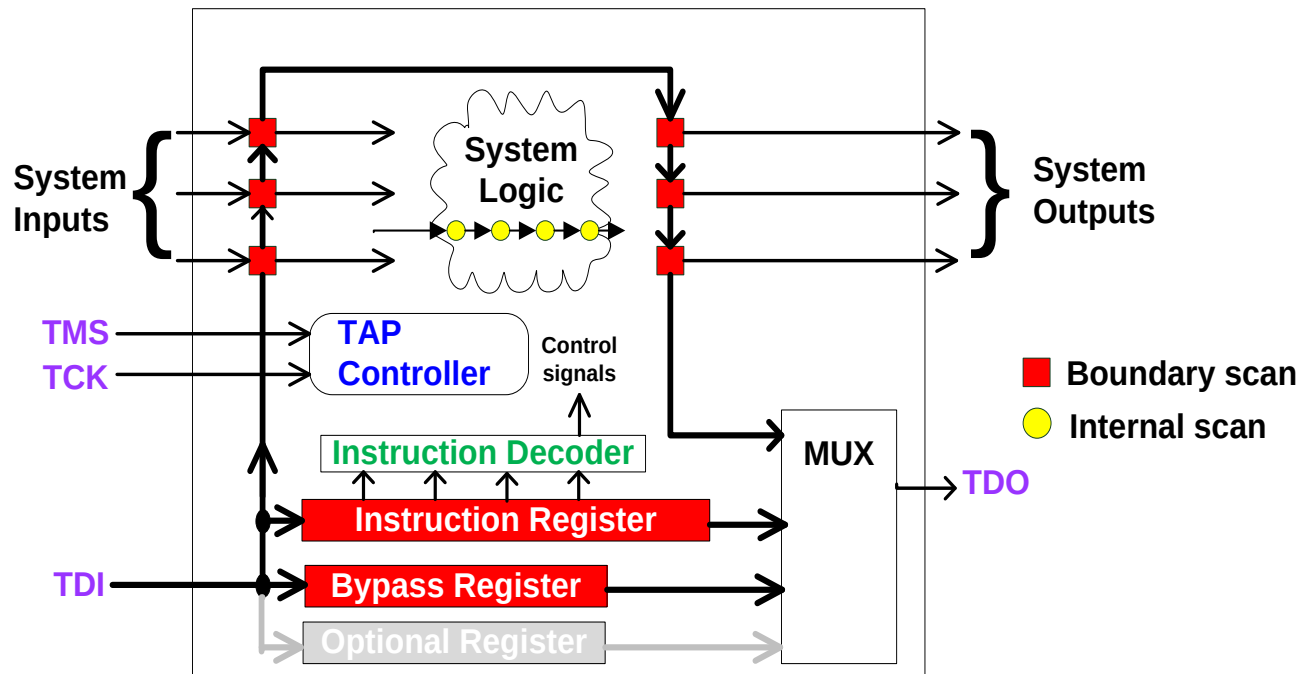
- ◆ 1. Test Access Port (TAP)

- ◆ 2. TAP controller

- ◆ 3. Registers

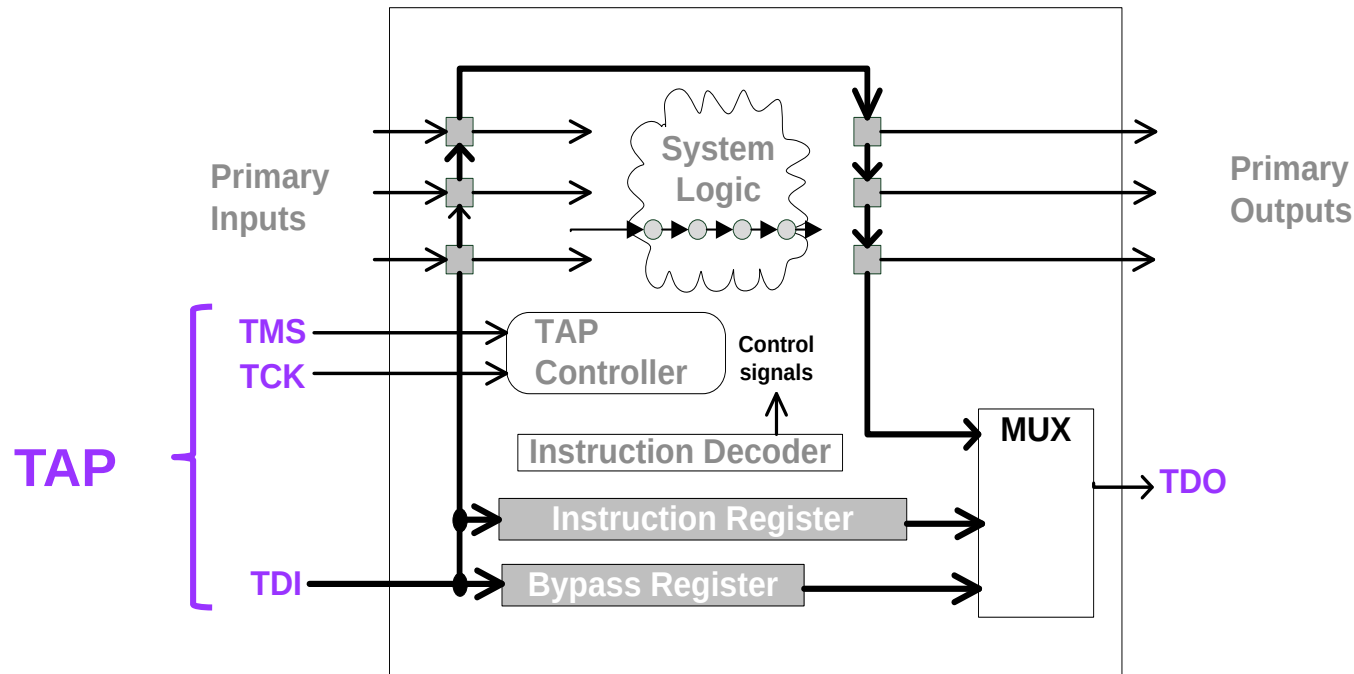
- \* Instruction Register, Boundary Scan Register, Bypass Register...

- ◆ 4. Instruction Decoder



# Test Access Port, TAP

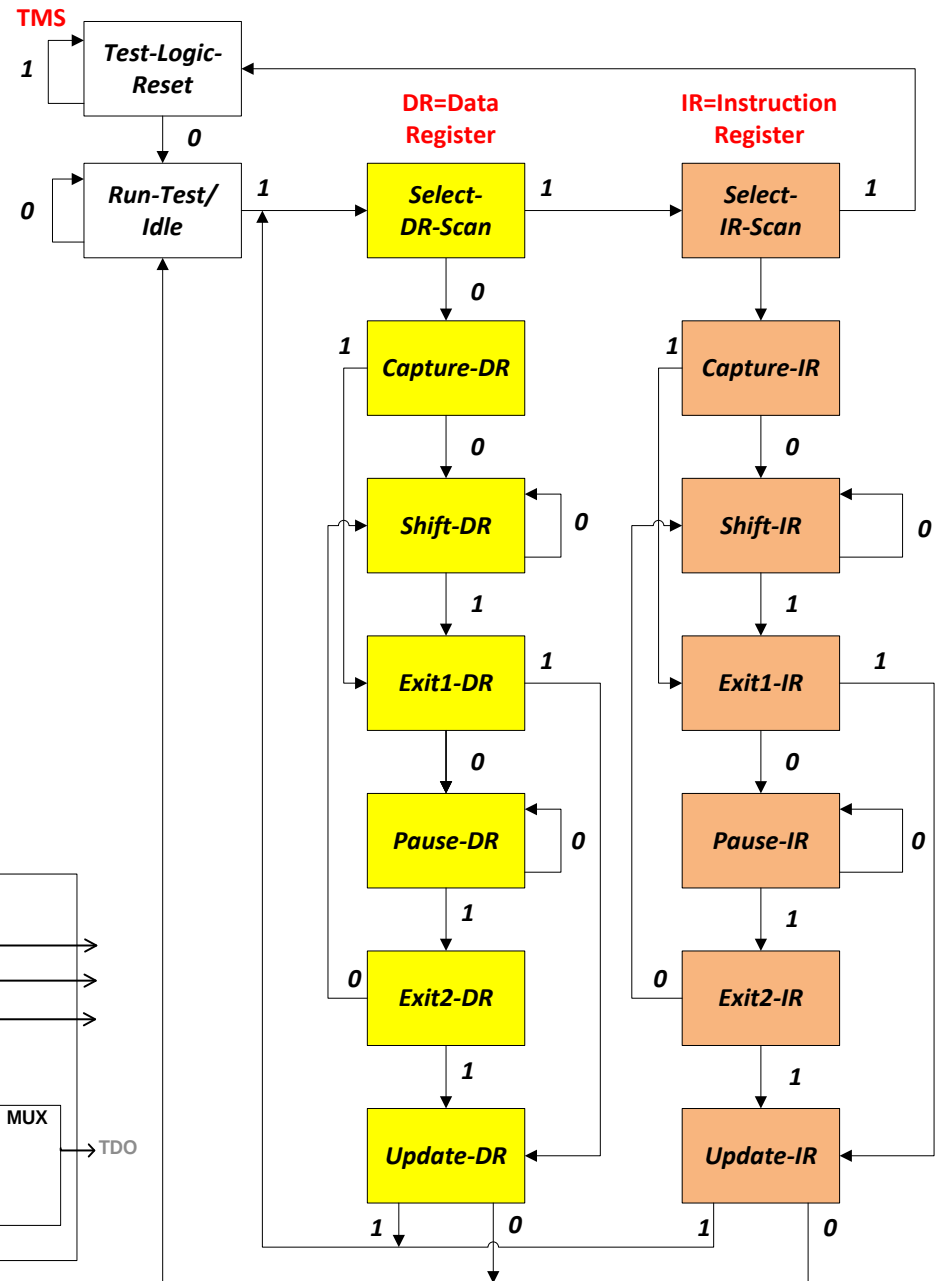
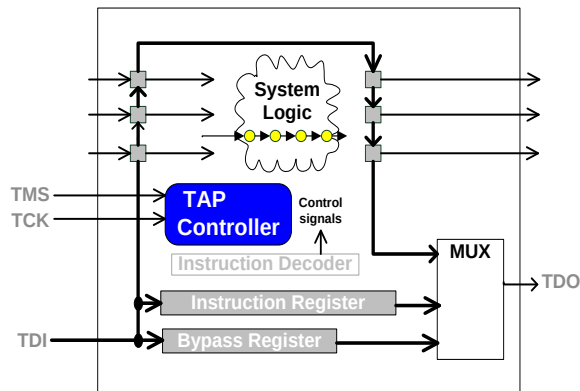
- 4 mandatory TAP
  - ♦ **TDI, Test Data Input**
  - ♦ **TDO, Test Data Output**
  - ♦ **TCK, Test Clock**
  - ♦ **TMS, Test Mode Select**
- 1 optional TAP
  - ♦ **TRST, Reset of Test Logic (Active Low)**



# TAP Controller

- Control JTAG operation
- 16-state Finite State Machine
  - ◆ Clock is TCK
  - ◆ Input is TMS
- Test-Logic-Reset
  - ◆ Reset JTAG Circuits
  - ◆ How to reset JTAG?
    - \* TMS=111..., or
    - \* TRST = 0

Note:  
State transition  
occurs at rising  
edge of TCK



# Summary

- **IEEE 1149.1-1990**
  - ♦ **Boundary scan** standard
  - ♦ aka **JTAG** (*Joint Test Action Group*)
- Why use boundary scan?
  - ♦ **Board-level, interconnect, system logic**
- JTAG components
  - ♦ 1. **Test Access Port (TAP)**
  - ♦ 2. **TAP controller** (16-state FSM)
  - ♦ 3. Registers
  - ♦ 4. Instruction Decoder



# FFT

- Q: What is mini number of 1's needed to initialize JTAG?  
Regardless of initial state.
  - ♦ TMS=111...

